Explaining and Predicting Oil Prices Using OPEC Communication

WIRTSCHAFTS UNIVERSITÄT WIEN VIENNA UNIVERSITY OF ECONOMICS AND BUSINESS

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Motivation and Research Question

- **Why forecast oil markets?**
  - Oil prices matter for inflation, growth, financial markets, energy policy

- **Challenge:** Prices react to new information & expectations → hard to predict

- **Role of OPEC:**
  - Announces production decisions & policy guidance.
  - Markets closely follow OPEC speeches & press releases.

- **Research question:**

Do OPEC communications contain information that helps us explain and predict oil prices, beyond standard numerical fundamentals?

Framework & Approach

- **Channel:** OPEC communication revises expectations about
  - future production, cooperation with non-OPEC, geopolitical risks

  → These expectation changes move futures prices, spot prices, inventories, production

- **Our approach:**
  - Extract text from speeches & press releases.
  - Turn it into high-dimensional text features.
  - Combine with numerical oil market fundamentals.
  - Use LASSO and other models to see whether the text adds predictive power.

Literature: OPEC Announcements & Oil Supply News

Känzig (2021, AER) – Oil Supply News Shocks

- Uses high-frequency changes in oil futures around OPEC announcements as an external instrument in a VAR.
- Identifies oil supply news shocks:
- Negative news (expected lower future supply) → immediate oil price increase, fall in production, higher inventories.
- Explain a meaningful share of oil price variation over time.

Relevance for us

- Confirms: OPEC announcements contain forward-looking information.
- We similarly treat OPEC communication as a signal about future supply

Literature II: OPEC Narratives & Text

Brunetti et al. (2024) – "Reasons Behind Words: OPEC Narratives and the Oil Market"

- Use text methods to build **narrative measures** from OPEC communication.
- Show that some narratives have only small direct effects on prices, but they help identify which type of shock (demand, supply, expectations) is driving the market.

Why this matters for us

- Shows that OPEC text can be measured systematically, not just read qualitatively.
- Motivates our approach: use bag-of-words + LASSO and let the data pick which words/phrases carry information.

Textual Data & Corpus

Textual data:

OPEC **speeches** (2003–2025)

- Scraped from yearly archive pages on opec.org.
- After cleaning: about **383** unique speeches.

OPEC **press releases** (2003–2025)

- Scraped from press section.
- After cleaning: about **572** unique releases.

Numerical Data & Transformation

Sources: U.S. Energy Information Administration, Yahoo Finance

Variables:

- Spot prices: WTI Crude Oil, CBOE Volatility Index, US Dollar Index
- Fundamentals: US field production, net imports, refinery utilization, total product supplied, inventories (crude oil, gasoline, distillate), strategic petroleum reserve

Transformations:

- Weekly fundamental data is propagated forward to daily rows (LOCF)
- Data is filtered to exclude weekends
- WTI Crude Oil and DXY are transformed into Daily Log-Returns
- VIX is used as a raw level proxy for market fear
- Fundamental data is transformed into 5-day percent changes (one week)

Text Collection (Scraping Pipeline)

Sources: OPEC press releases + speeches from official website (2003–2025)

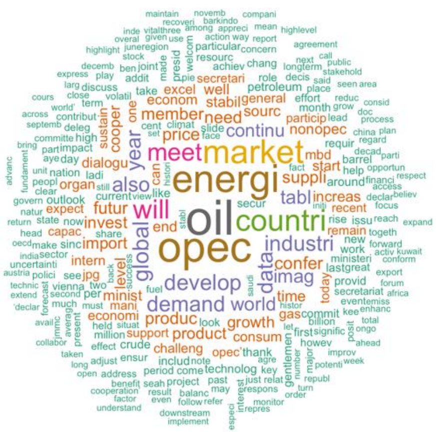
Scraping: For each year-page, collect links to individual documents and download

Extracted info:

- date
- main text (paragraphs and tables)
- URL

Text Descriptive and Wordclouds

OPEC Text — Wordcloud (Unigrams)

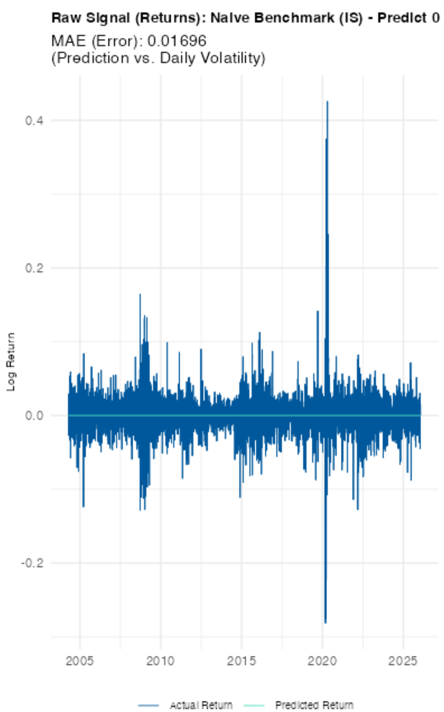


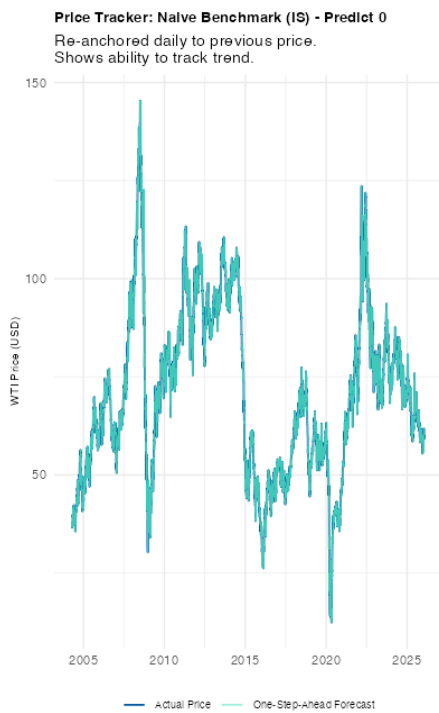
OPEC Text — Wordcloud (Bigrams)

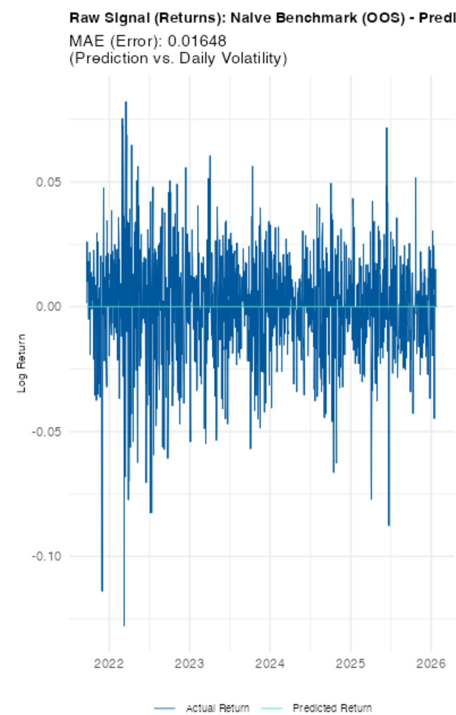
Naïve model

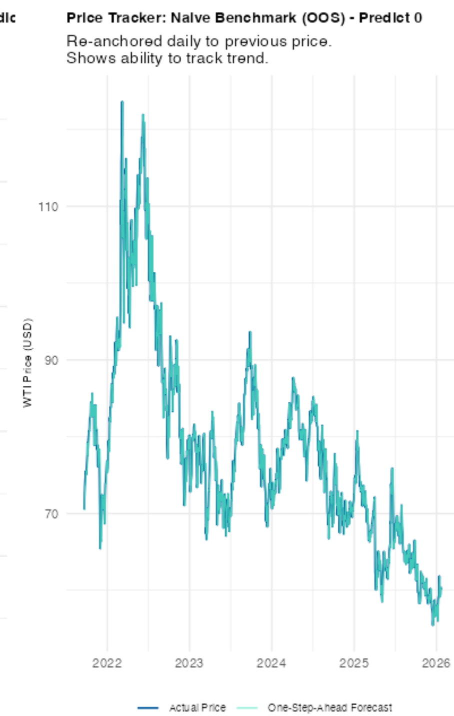
- **Assumption:** Market is efficient and current prices already reflect all available public information. Therefore, the expected return is zero.
- **Implication**: Oil price follows a random walk with no drift, a stochastic process where the conditional expectation of the next value, given all prior values, is equal to the present value.
- **Rule:**
    - Price assumption: $E[P_{t+1}|P_t] = P_t$
    - Return assumption: $E[r_{t+1}] = 0$
- **Performance:** MAE ≈ 0.016–0.017 (about the size of daily volatility)
- **Takeaway:** Any “smart” model (LASSO) must beat this error, especially out-of-sample

Naïve model


Raw Signal (Returns): Naive Benchmark (IS) - Predict 0
MAE (Error): 0.01696
(Prediction vs. Daily Volatility)
Log Return
0.4
0.2
0.0
-0.2
2005
2010
2015
2020
2025
Actual Return
Predicted Return


Price Tracker: Naive Benchmark (IS) - Predict 0
Re-anchored daily to previous price.
Shows ability to track trend.
WTI Price (USD)
150
100
50
2005
2010
2015
2020
2025
Actual Price
One-Step-Ahead Forecast


Raw Signal (Returns): Naive Benchmark (OOS) - Predic
MAE (Error): 0.01648
(Prediction vs. Daily Volatility)
Log Return
0.05
0.00
-0.05
-0.10
2022
2023
2024
2025
2026
Actual Return
Predicted Return


Price Tracker: Naive Benchmark (OOS) - Predict 0
Re-anchored daily to previous price.
Shows ability to track trend.
WTI Price (USD)
110
90
70
2022
2023
2024
2025
2026
Actual Price
One-Step-Ahead Forecast

LASSO – Model Explanation

What we do

- Feature selection: drives many β_j to 0 → picks a small set of words
- Handles high-dimensional text: works when $p \gg n$
- Interpretability: non-zero coefficients show which terms matter

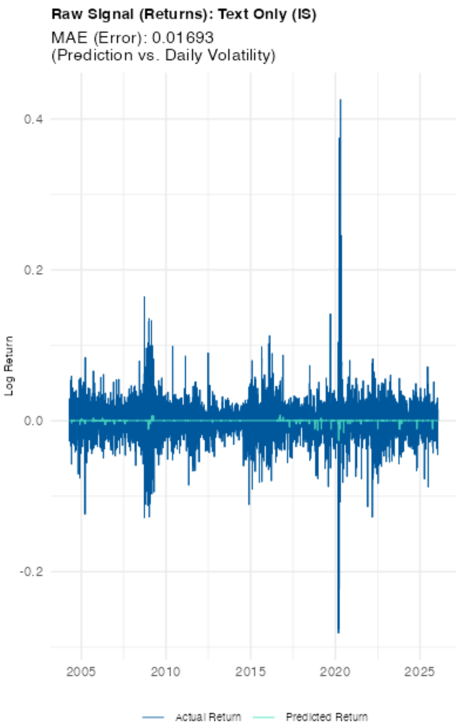
Model Objective

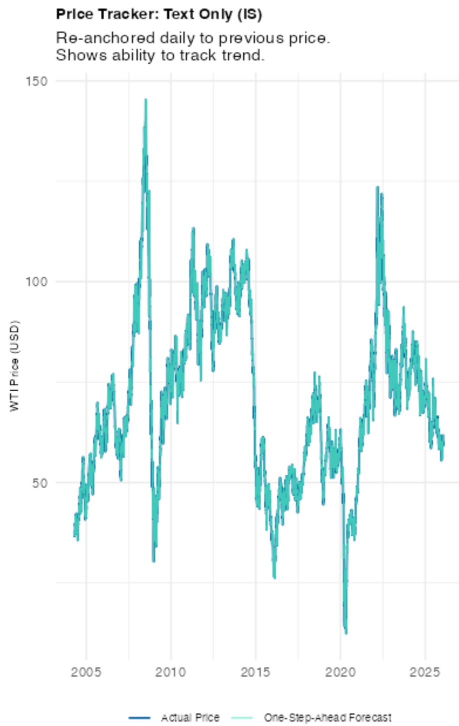
$$\hat{\beta} = \arg\min_{\beta} \left\{ \frac{1}{n} \sum_{t=1}^{n} (y_t - \beta_0 - x_t^T \beta)^2 + \lambda \sum_{j=1}^{p} |\beta_j| \right\}$$

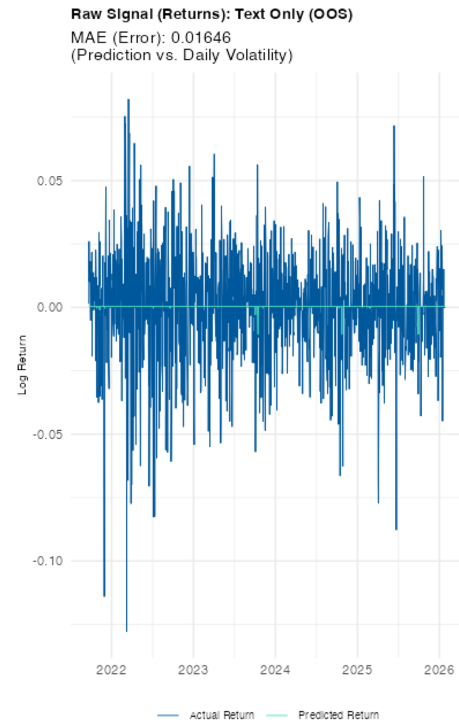
Key Explanations

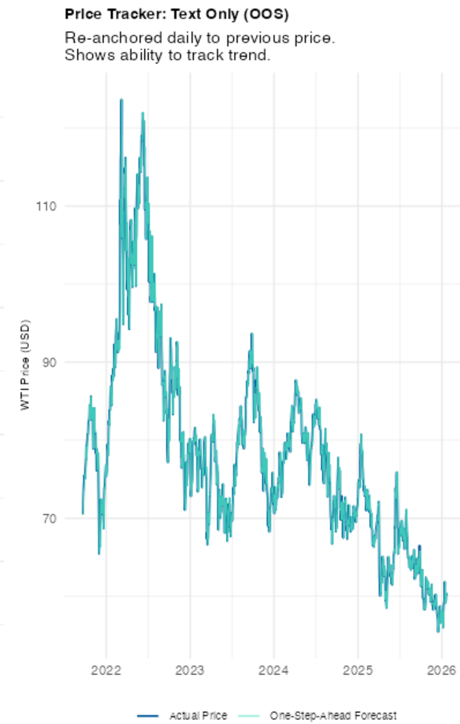
- y_t: Daily WTI crude oil return
- $x_t^T \beta$: Combined signal of OPEC textual features and market fundamentals
- λ: The "Key Knob" controlling shrinkage; larger λ results in fewer non-zero coefficients

Lasso (text only)

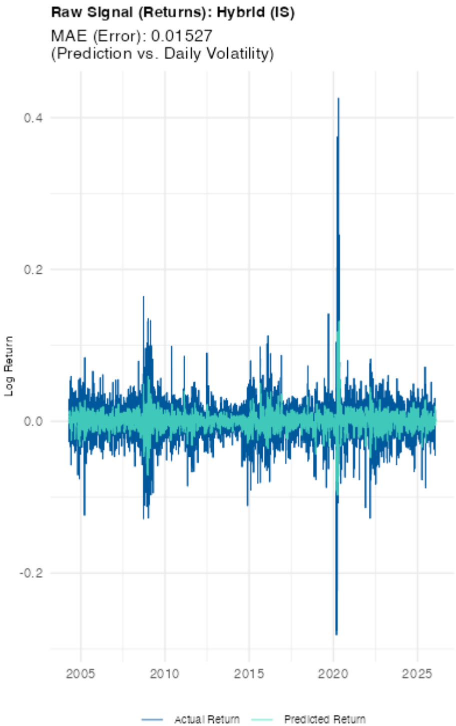


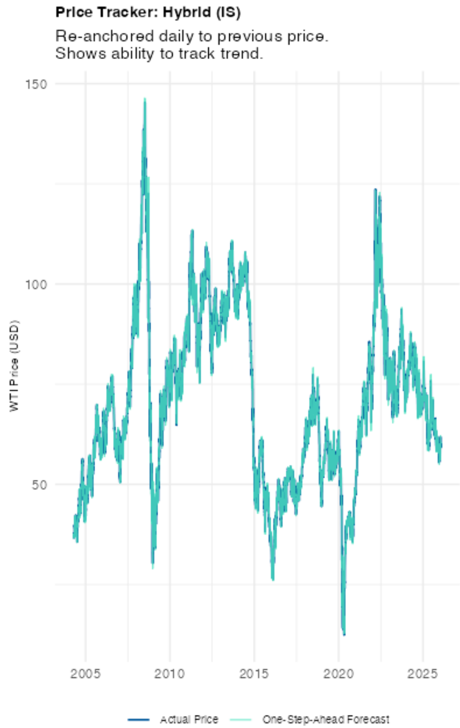


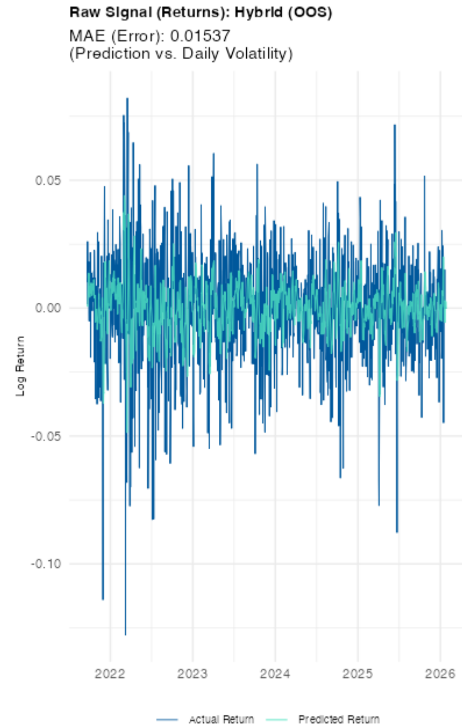


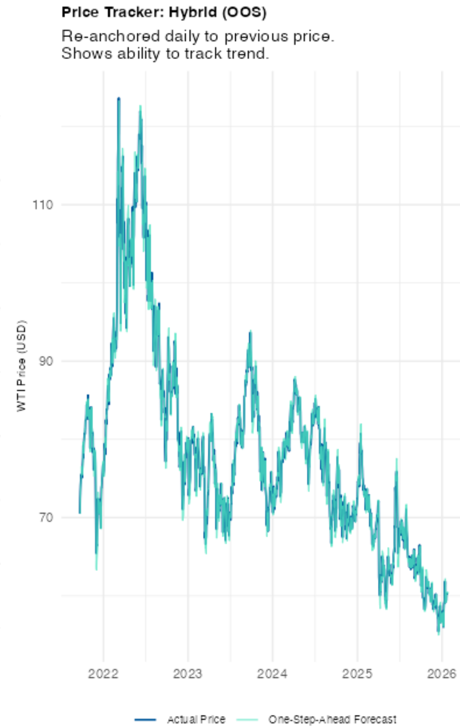


Lasso hybrid









Elastic Net – Model Explanation

What we do

- LASSO + Ridge: combines selection (L1) and stability (L2)
- Collinearity-friendly: keeps groups of correlated words together
- Often more stable than pure LASSO with text features

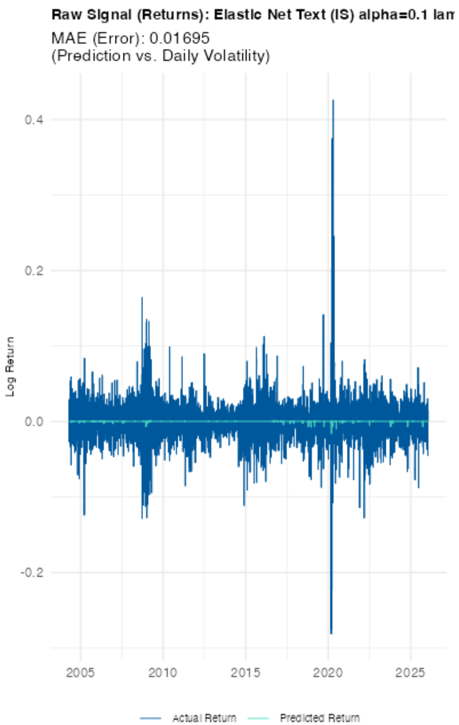
Model Objective

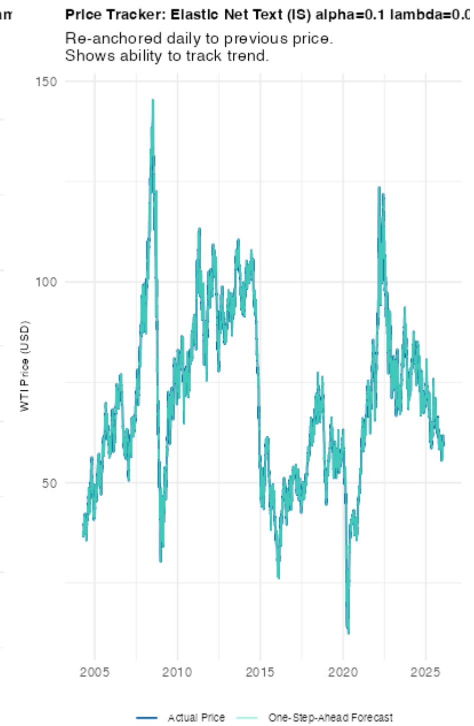
$$\hat{\beta} = \arg\min_{\beta} \left\{ \frac{1}{n} \sum_{t=1}^{n} (y_t - \beta_0 - x_t^T \beta)^2 + \lambda \left[\alpha \sum_{j=1}^{p} |\beta_j| + (1-\alpha) \frac{1}{2} \sum_{\mathrm{j}=1}^{\mathrm{p}} \beta_{\mathrm{j}}^2 \right] \right\}$$

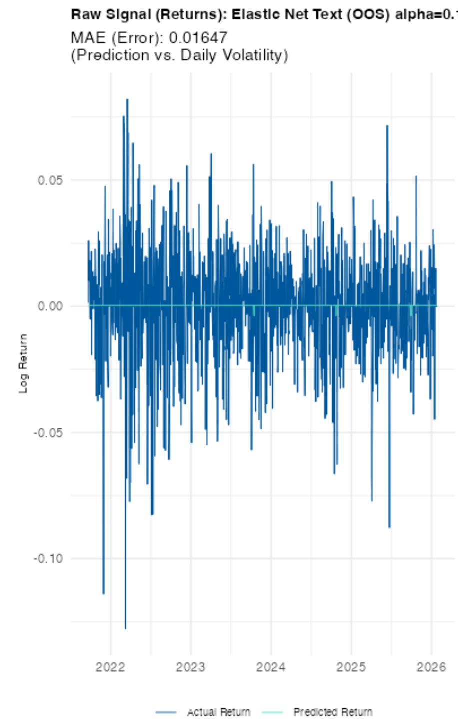
Key Explanations

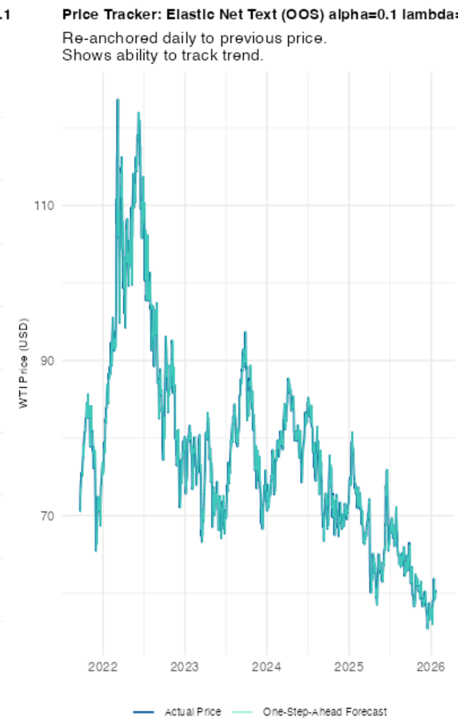
- α: The mixing parameter; $\alpha = 1$ is LASSO, while $\alpha = 0$ is Ridge
- Cross-Validation: We tuned α values (0.1, 0.5, 0.9) using CV to respect the time-series structure
- λ: Controls the overall strength of regularization

Elastic Net (text only)

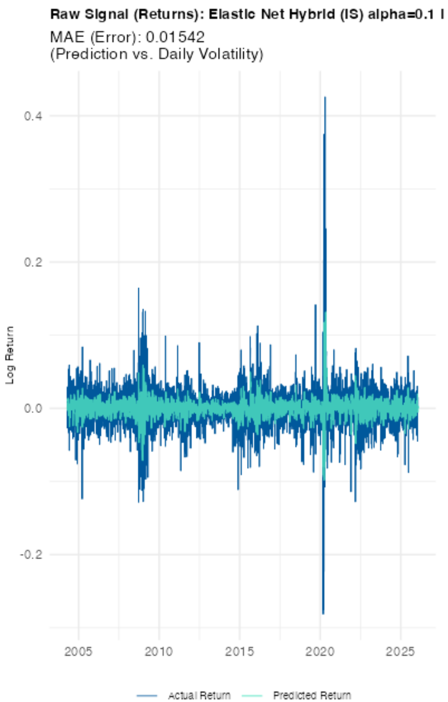


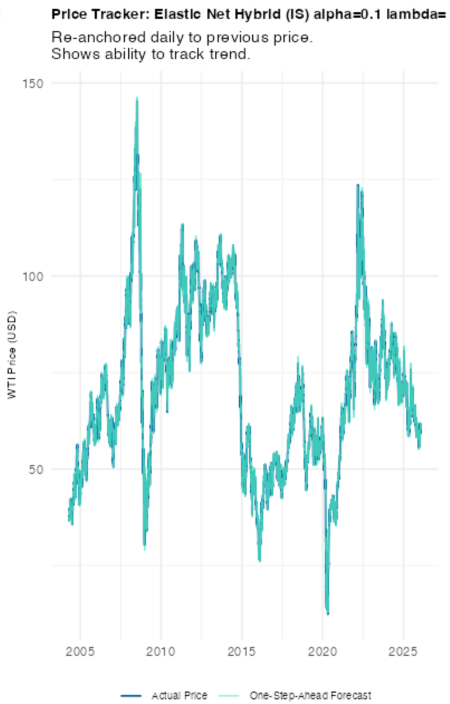


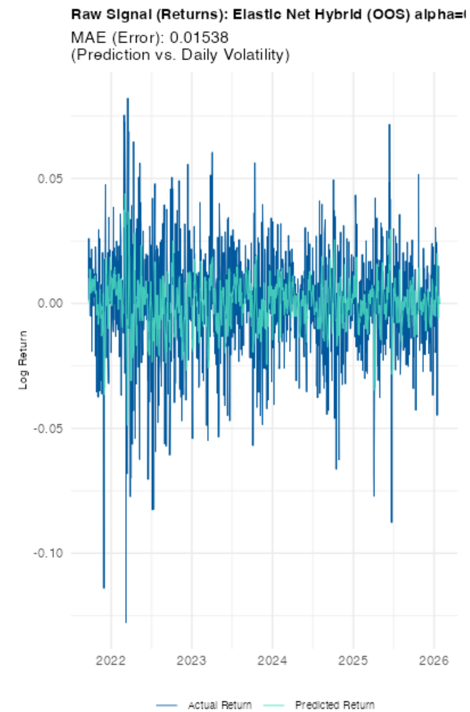


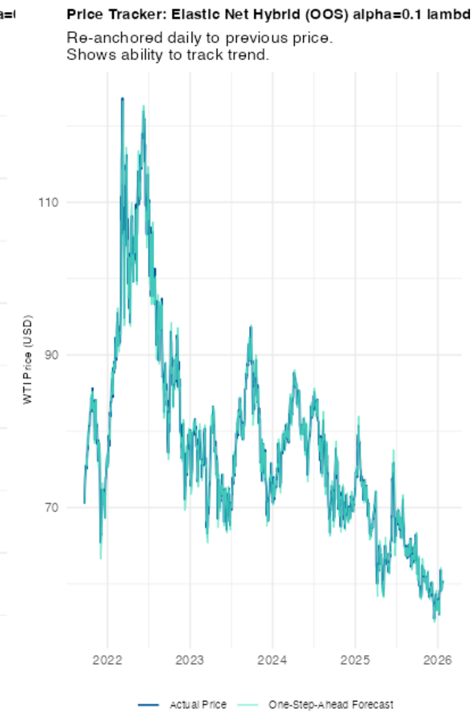


Elastic Net (hybrid)









Random Forest

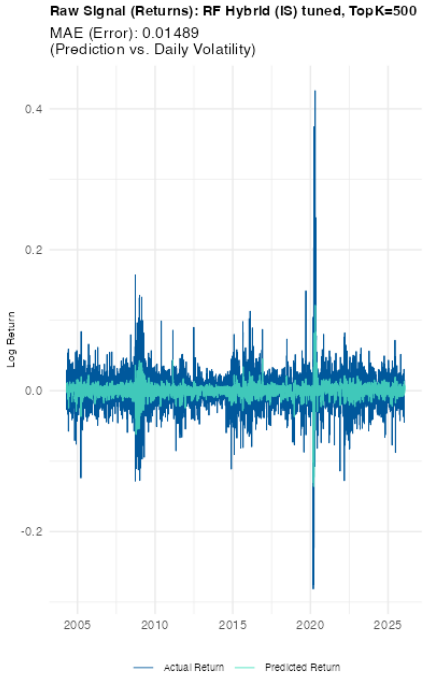
- We use bootstrap aggregating to construct B uncorrelated decision trees – at each split, only a random subset of features is considered
- The final forecast is the average of all individual tree predictions
- Reduces variance and is robust against overfitting, especially with high-dimensional text data (sparse DTM)
- Hyperparameter tuning performed via internal time-series cross-validation (no future data leakage)

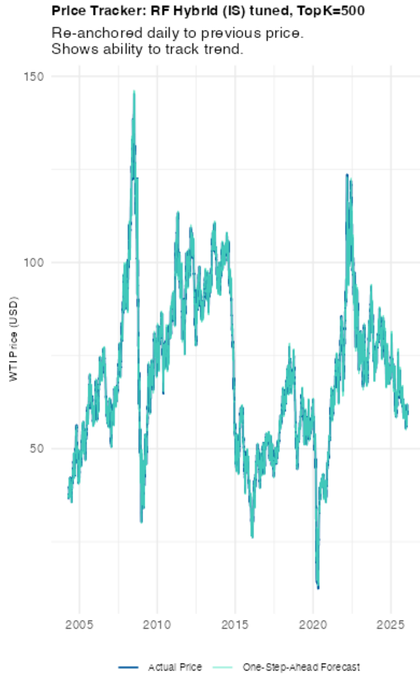
For a given input vector x (numerical + textual data), the prediction $\hat{y}$ is:

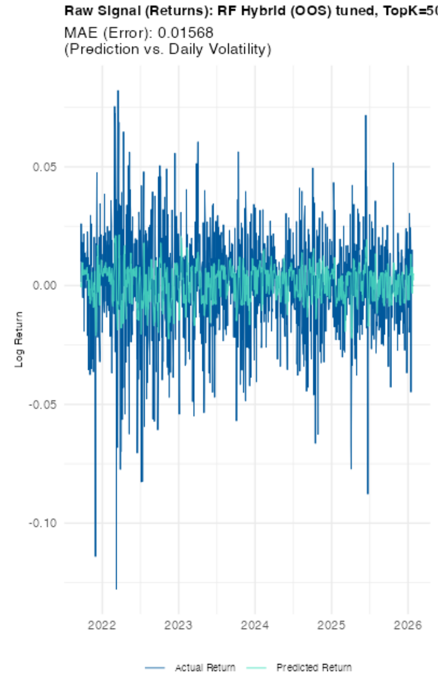
$$\hat{y} = \frac{1}{B}\sum_{b=1}^{B} T_b(x)$$

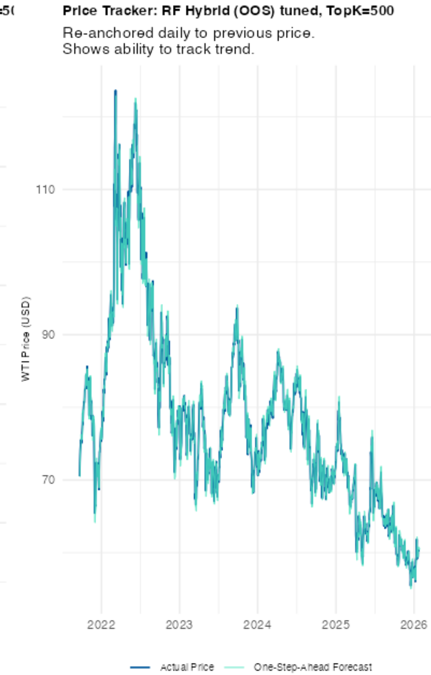
where $T_b(x)$ is the prediction of the b-th tree and B is the number of trees

Random Forest









Time-series cross-validation selected a highly regularized configuration (Depth=12, MinNode=20) with a large ensemble of 2,000 trees and a 500-word vocabulary to stabilize predictions against noisy text data.

XGBoost (Gradient Boosting)

- Unlike random forest, XGBoost builds trees sequentially – each new tree corrects the residual errors of the previous ensemble
- Minimizes a regularized objective function using gradient descent in function space
- Includes L1/L2 regularization terms (Ω) to control model complexity and prevent overfitting on noise

The prediction at step t is the sum of the previous prediction plus the new tree f_t, scaled by the learning rate η:

$$\hat{y}_i^{(t)} = \hat{y}_i^{(t-1)} + \eta f_t(x_i)$$

$$Objective\colon \mathcal{L}(\theta) = \sum_i l(y_i, \hat{y}_i) + \sum_k \Omega(f_k)$$

XGBoost

The model was tuned for maximum robustness, selecting a very conservative learning rate (η=0.01), shallow trees (Depth=3), and a strict 200-word vocabulary to prevent overfitting on the sparse textual features.

Overview of Results

Mean Absolute Error	Naïve	LASSO	Elastic Net	Random Forest	XGBoost
Text Only (in-sample)	0.01696	0.01693	0.01695	0.01681	**0.01654**
Hybrid (in-sample)	0.01696	0.01527	0.1542	**0.01489**	0.01499
Text Only (out-of-sample)	0.01648	**0.01646**	0.01647	0.01649	0.01649
Hybrid (out-of-sample)	0.01648	**0.01537**	0.01538	0.01568	0.01546

- Combining text and macro data consistently beats the Naïve benchmark, proving sentiment adds predictive value.
- The linear LASSO Hybrid model (MAE 0.01537) outperforms complex non-linear models (RF, XGBoost) on unseen data.
- The best model reduces forecast error by ~6.7% compared to the market benchmark.
- While Random Forest fits training data best (potential overfitting), LASSO generalizes best to the test set.

Limitations and Future Work

Limitations

- Models predict only t+1, we did not test the ability to forecast longer-term trends
- Document-term matrix ignores word order and context, and might therefore lose semantic nuance compared to modern NLP
- Speeches and press releases are sporadic – on days without events, the text signal is zero

Future Research

- Using the monthly OPEC oil market report, the monthly EIA short term energy outlook or the EIA weekly patroleum status report as alternative/additional data sources
- Test predictive power over longer horizons
- Focus on intraday volatility impact in the minutes/hours following a speech/press release

Conclusion

- Combining textual sentiment with macroeconomic fundamentals consistently outperforms the Naïve benchmark, reducing the out-of-sample forecast error by ~6.7% (LASSO Hybrid).
- Quantitative fundamentals remain the primary drivers of oil price trends, with Momentum (Lagged Returns), Volatility (VIX), and Currency Strength (DXY) identified as the strongest predictors across all models.
- Qualitative text features—specifically those related to OPEC events (e.g., 'secretariat', 'vienna')—rank within the top predictors, providing a distinct "fine-tuning" signal that captures market reactions to specific policy announcements.
- The linear LASSO model achieved the highest out-of-sample accuracy, outperforming complex non-linear models (Random Forest, XGBoost), suggesting that strict feature selection is more effective than model complexity in noisy financial environments.
- The ability to systematically beat the random walk suggests that the oil market is not fully efficient and contains exploitable information asymmetries embedded in qualitative communications.

Thank you for your attention!

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